UNIVERSITY OF SINDH

JAMSHORO SINDH, PAKISTAN

B. Ed. 04 Year

Course No. GICT-300 Credit Hours **3**

Title of Course **Applications of Information and Communication Technology (ICT)** Semester: 1st Semester

Course Description

This course provides a foundational understanding of Information and Communication Technology (ICT) and its applications for students across arts, sciences, and social sciences disciplines. It introduces fundamental concepts of computer literacy, computer systems, hardware and software components, operating systems, internet usage, cybersecurity, and ethical issues related to digital technologies. The course emphasizes practical use of ICT tools for academic, professional, and everyday purposes, while also exploring emerging technologies and their impact on society and future careers.

Learning Outcomes

- Explain the concept and importance of computer literacy in modern society.
- Classify computers based on functionality and size.
- Identify major hardware and software components of computer systems.
- Describe basic operating system concepts.
- Apply safe and responsible practices for cybersecurity and privacy.
- Demonstrate ethical use of ICT and avoid plagiarism.
- Use internet tools and digital communication platforms effectively.
- Explain emerging technologies and their societal impact.

Course Outline

Unit 1: Introduction to Computer Literacy and ICT

- Concept and scope of computer literacy
- Importance of ICT in education, work, and society
- Evolution and generations of computers
- Classification of computers
- Impact of computers and the internet on society
- Applications of ICT in various fields

Unit 2: Computer Hardware and Software Fundamentals

- Input, output, and storage devices
- Primary and secondary memory
- System software and operating systems
- Application software and productivity tools
- Communication hardware and software

Unit 3: Operating System Concepts

- Functions and types of operating systems
- File and process management
- Input/output management
- Kernel functions

Unit 4: Cyber security and Data Protection

- Cyber threats and online security
- Digital footprints
- Password management and authentication
- Cloud computing basics
- Privacy and data protection issues

Unit 5: Ethical and Responsible Use of ICT

- Digital citizenship
- Online etiquette
- Intellectual property rights
- Avoiding plagiarism

Unit 6: Internet Applications and Emerging Technologies

- Web browsers and search engines
- Online communication tools
- Safe browsing practices
- Artificial Intelligence

- Blockchain
- Internet of Things (IoT)
- Virtual and Augmented Reality

Teaching / Instructional Strategies

1. Lectures
2. Interactive discussions
3. Question–answer sessions
4. Demonstrations
5. Case study analysis

Practical Work

6. Hands-on computer exercises
7. Individual assignments
8. Group assignments
9. Internet research tasks
10. Presentations

Recommended Books / References

- Shelly, G. B., & Vermaat, M. E. Discovering Computers. Cengage Learning.
- Norton, P. Introduction to Computers. McGraw-Hill Education.
- Stair, R., & Reynolds, G. Principles of Information Systems. Cengage Learning.
- Laudon, K. C., & Laudon, J. P. Management Information Systems. Pearson.
- Higher Education Commission (HEC), Pakistan. ICT and Digital Literacy Guidelines.